

Article

PID Sliding Mode Control of PMSM Based on Improved Terminal Sliding Mode Reaching Law

Guodong Qin ¹, Min Wang ², Guizhou Cao ³ , Qi Wang ^{4,*} and Yuefeng Liao ²¹ School of Intelligent Manufacturing, Jiaxing Vocational and Technology College, Jiaxing 314036, China² School of Electrical and Information Engineering, Zhengzhou University, Zhengzhou 450001, China³ State Grid Henan Electric Power Research Institute, Zhengzhou 450001, China⁴ Institute of Scientific and Technical Information, Zhengzhou University, Zhengzhou 450001, China

* Correspondence: qwang.cn@outlook.com

Abstract: In order to enhance the dynamic performance and anti-disturbance ability of speed control for a permanent magnet synchronous motor (PMSM), a sliding mode control method based on a PID sliding surface and an improved terminal sliding mode reaching law (ITSMRL) is proposed. Firstly, an ITSMRL is proposed to increase the reaching speed and reduce chattering; moreover, it has been verified that the reaching law (RL) can achieve a sliding mode surface in finite time. Then, based on the dynamic model of PMSMs with uncertainties, an extended state observer (ESO) is used to estimate the lumped disturbance, and it is proven that the estimated error is bounded. Finally, on the basis of the observed feedforward disturbance, to enhance the disturbance rejection ability of PMSMs, a controller that combines the PID sliding mode surface and the ITSMRL is proposed. Moreover, the stability of the closed-loop system is proven. The composite method has the characteristics of a fast reaching speed, small chattering and strong robustness, and is verified by experiments.

Keywords: PMSM; improved reaching law; anti-disturbance ability



Academic Editor: Ahmed Abu-Siada

Received: 28 February 2025

Revised: 21 April 2025

Accepted: 14 May 2025

Published: 21 May 2025

Citation: Qin, G.; Wang, M.; Cao, G.; Wang, Q.; Liao, Y. PID Sliding Mode Control of PMSM Based on Improved Terminal Sliding Mode Reaching Law. *Energies* **2025**, *18*, 2661. <https://doi.org/10.3390/en18102661>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

PMSMs have the advantages of small size, high power density, and large output torque. Owing to these merits, they are widely utilized in industrial automation, electric vehicles, wind power generation, household appliances and other fields [1–3]. For the control of PMSMs, the traditional methods include proportional–integral (PI) control [4], variable structure control (VSC) [5], model predictive control (MPC) [6], active disturbance rejection control (ADRC) [7], fuzzy control (FC) [8], etc. In recent years, with the development of artificial intelligence, edge computing and advanced control theory, a variety of new control strategies have emerged in PMSM control systems, such as data-driven control [9], mode-switching control [10], reinforcement learning optimization control [11], event-triggered control [12], etc. Among these, variable structure control has been widely used in the field of PMSM control because of its robustness to external perturbation, fast response and low requirements for system parameter accuracy [13–17]. Sliding mode control (SMC) is synonymous with VSC. Due to the existence of the signum function, chattering exists in SMC. Moreover, this chattering can only be weakened to a certain extent, or the disturbance rejection capacity of the sliding mode will be eliminated. Serious chattering will stimulate the dynamic properties of the omitted high-frequency characteristics, affect the control accuracy of the controller, enhance the energy consumption of the system, and potentially result in system vibrations and damage to the hardware equipment of the control system.

Therefore, reducing chattering is the primary research direction related to SMC. In recent years, various methods have been proposed to solve the problem of chattering in PMSM systems, such as the reaching law (RL) method [18], disturbance observer composite method [19], filter method [20], boundary layer method [21], sliding mode control with fractional order characteristics [22], etc.

The disturbance observer composite method can effectively reduce the gain coefficient of a sliding mode controller and further reduce chattering by feeding the observed disturbance forward. Given that extended state observers (ESOs) estimate both system states and disturbance dynamics not based on a system model, and the ease of adjusting the parameters, they are often used to design composite controllers combined with the sliding mode control method [23–26]. Ma et al. designed a compound integral terminal sliding mode control (ITSMC) in discrete form with an ESO for the purpose of PMSM speed control [23]. A combined approach which integrated continuous fast terminal sliding mode control (FTSMC) and an ESO was presented in [24]. A speed-regulating controller based on proportional integral sliding mode control (PISMC) and an ESO was developed in [25]. A composite speed control method has been put forward by Liu et al., featuring a modified super-twisting SMC and an ESO [26]. However, the above composite methods of SMC and ESO are all based on conventional RLs.

The RL should give consideration to not only fast reaching speed but also small chattering; that is, the RL is large when the condition of the system shows that its state is distant from a sliding mode surface, and the RL is small when the system state is near a sliding mode surface. The exponent RL was first designed by Gao et al., which is faster than the constant rate RL [27]. Given that the system state is far from the sliding mode surface, the power RL is greater than the constant rate RL. Conversely, when the system state approaches a sliding mode surface, the power RL is smaller than the constant rate RL [28]. Xu et al. designed a terminal sliding mode RL (TSMRL), which showed better performance than the power RL in terms of fast reaching speed and small chattering [24]. Wang et al. advanced the conventional power RL by integrating system state variables and an exponential function, which alleviates chattering and reduces the time required to reach the desired state [29]. However, the law contains six parameters. Therefore, an effective RL should not only balance a reduction in chattering with the acceleration of convergence but also give consideration to the complexity inherent in the algorithm structure.

What is more, in order to improve tracking performance and enhance disturbance rejection ability, PID sliding mode surfaces have been designed for some application situations [30–32], especially PMSMs [33,34]. A composite fractional-order PID sliding mode controller with an ESO was designed for the speed control of a surface-mounted permanent magnet synchronous motor (SPMSM) [33]. A composite adaptive super-twisting controller with a nonlinear fractional-order PID sliding mode surface and ESO for PMSM speed control was proposed in [34]. In recent years, control methods involving fractional-order sliding mode composite disturbance observers, particularly terminal sliding mode control (TSMC) and non-singular terminal sliding mode control (NTSMC), have been applied to the speed control systems of permanent magnet synchronous motors (PMSMs). However, the observers used are always asymptotically stable and cannot guarantee the finite-time stability of a closed-loop system [14,24,35,36]. Moreover, due to the large number of parameters involved, a fractional-order sliding mode surface will face complexities when adjusting the parameters.

Motivated by the aforementioned papers, in order to balance the reduction in chattering with the acceleration of convergence and enhance the disturbance rejection ability, we propose a novel controller of PMSMs based on a PID sliding mode surface with an improved terminal sliding mode RL (ITSMRL) and an ESO. In particular, the ITSMRL is better than the TSMRL for achieving fast reaching speed and reducing chattering.

Compared with previous relevant studies, the major contributions of this paper are as follows:

- (1) An ITSMRL is proposed, which not only balances the reduction in chattering with the acceleration of convergence but also considers the complexity of the structure. It has been verified that the RL can reach a sliding mode surface within a finite time.
- (2) A novel speed composite controller based on a PID sliding mode surface with an ITSMRL and ESO is designed. The boundedness of the estimated error of the ESO and the stability of the closed-loop system are proven.
- (3) The validity and robustness of the proposed algorithm are evidenced by simulation and experimental tests.

The remainder of this paper is presented as follows. Section 2 presents a mathematical model with the uncertainty of an SPMSM. In Section 3, an ITSMRL is designed and its finite reaching time to the sliding mode surface is proven. Moreover, a PID sliding mode controller for the SPMSM based on an ITSMRL and ESO is also presented. The simulated and experimental results are presented in Section 4. Finally, in Section 5, the conclusions are presented.

2. Preliminaries

In this section, a dynamic equation of a PMSM in a speed loop with uncertainty is given. The following statement elaborates the mathematical model of the PMSM [37–40]:

$$\begin{cases} L_d \frac{di_d}{dt} = -R_s i_d + n_p \omega L_q i_q + u_d, \\ L_q \frac{di_q}{dt} = -R_s i_q - n_p \omega L_d i_d - n_p \omega \Phi + u_q, \\ J_m \frac{d\omega}{dt} = \tau - \tau_L - B\omega, \end{cases} \quad (1)$$

where i_d , i_q , u_d , u_q and L_d , L_q are the d - q axis current, d - q axis voltage, and d - q axis stator inductances, respectively. n_p is the number of pole pairs, Φ is the rotor flux linkage, R_s is the stator resistance, J_m is the inertia, ω is the mechanical angular velocity, τ_L is the load torque and B is the viscous friction coefficient. The formula of electromagnetic torque is $\tau = \frac{3}{2} n_p [(L_d - L_q) i_d i_q + \Phi i_q]$, where $L_d = L_q$ in the SPMSM.

The mathematical model of the uncertainty of the SPMSM in the speed loop can be described as

$$\dot{\omega} = a_e + d_\omega, \quad (2)$$

where d_ω is the lumped disturbance, including parameter variations, unknown load torques, unmodeled dynamics and external disturbances:

$$\begin{cases} a_e = \frac{3n_p \Phi}{2J_m} i_q, \\ d_\omega = -\frac{\tau_L}{J_m} - \frac{B\omega}{J_m} + \frac{\frac{3}{2} n_p \Delta \Phi i_q - \Delta J_m \omega}{J_m} + w_\omega, \end{cases} \quad (3)$$

where $\Delta \Phi$ and ΔJ_m are variations in the Φ and J_m , respectively. In addition, w_ω is the external disturbance.

Assumption 1. The lumped disturbance of (2) is bounded and continuously differentiable [24,29,37,41,42].

3. Design of Sliding Mode Controller Based on ESO and ITSMRL

Figure 1 illustrates the proposed control scheme for the PMSM system. The whole control block diagram includes three parts, which are the speed loop controller, current loop

controller and PMSM controlled-object. In the speed loop, firstly, the ITSMRL is proposed. Then, the ESO is developed with the purpose of estimating the total lumped disturbance. Finally, the observed lumped disturbance is fed forward; based on this, a composite sliding mode control method based on the ITSMRL is designed.

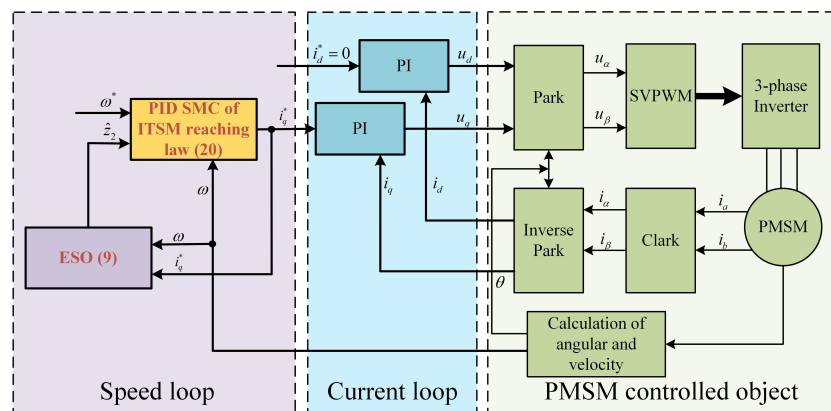


Figure 1. The control structure of a PMSM system with the proposed approach.

3.1. Improved TSMRL Design

In this section, the TSMRL is first presented, and then the, ITSMRL is proposed.

3.1.1. TSMRL

The RL should give consideration to fast approach and small chattering; that is, the reaching speed adapts based on distance to the sliding surface: higher when far away, lower when near.

The TSMRL is

$$\dot{s} = -k_1 |s|^{1-\beta} \text{sign}(s) - k_2 s, \quad (4)$$

where s is the sliding mode surface, $\lim_{s \rightarrow 0} \dot{s} \leq 0$, $k_1 > 0$, $k_2 > 0$ and $0 < \beta < 1$. The TSMRL can be thought of as the combination of the power RL $\dot{s} = -k_1|s|^{1-\beta}\text{sign}(s)$ and the exponent RL $\dot{s} = -k_1\text{sign}(s) - k_2s$. During the reaching phase, the system demonstrates accelerated convergence relative to the power RL.

3.1.2. Proposed ITSMRL

In order to further accelerate convergence when the state of the system is significantly distant from the sliding mode surface and reduce chattering when the system state is near the sliding mode surface, two power functions are introduced. The first power function is $|s|^{1+\beta}$, which enables the system to converge more rapidly when its state is substantially distant from the sliding mode surface. The second power function is $|x_1|^{1+\beta}$, where x_1 is the system state error and $\lim_{t \rightarrow \infty} x_1 = 0$. When the state of the system approaches the sliding mode surface, it becomes small.

In this paper, an improved TSMRL (ITSMRL) is proposed, which has faster reach and reduced chattering compared to the above TSMRL. The ITSMRL can be represented as

$$\dot{s} = -k_1|x_1|^{1+\beta}|s|^{1-\beta}\text{sign}(s) - k_2|s|^{1+\beta}\text{sign}(s), \quad (5)$$

where $k_1 > 0$, $k_2 > 0$, $0 < \beta < 1$. When $|s| > 1$ and $|x_1| > 1$, the ITSMRL $|-k_1|x_1|^{1+\beta}|s|^{1-\beta}\text{sign}(s) - k_2|s|^{1+\beta}\text{sign}(s)|$ is bigger than the TSMRL $|-k_1|s|^{1-\beta}\text{sign}(s) - k_2s|$. When $|s| < 1$ and $|x_1| < 1$, the ITSMRL is smaller than the TSMRL. Therefore, the ITSMRL has faster reaching speed and less chattering than the TSMRL.

Theorem 1. The proposed ITSMRL (5) will reach convergence in finite time.

Proof. Define $V = \frac{s^2}{2}$; according to (5), the derivative of V can be obtained as

$$\dot{V} = s\dot{s} = -k_1|x_1|^{1+\beta}|s|^{2-\beta} - k_2|s|^{2+\beta} \leq 0 \quad (6)$$

Because of $s\dot{s} \leq 0$, the existence and reachability conditions of the continuous sliding mode systems are satisfied, and there is a RL and an equilibrium point $s = 0, \dot{s} = 0$.

The reaching time can be obtained by integrating (5) as follows:

$$\begin{aligned} t &= \int_{|s(0)|}^0 \frac{1}{-k_1|x_1|^{1+\beta}|s|^{1-\beta} - k_2|s|^{1+\beta}} ds \\ &= \frac{\arctan \left[s(0)^{\beta} k_1^{-\frac{1}{2}} k_2^{\frac{1}{2}} |x_1|^{-\frac{1+\beta}{2}} \right]}{\beta k_1^{\frac{1}{2}} k_2^{\frac{1}{2}} |x_1|^{\frac{1+\beta}{2}}} \end{aligned} \quad (7)$$

The proof is thus completed. $\square$

3.2. Design of ESO

Define $x_1 = \omega$ and $x_2 = d_\omega$; consequently, the system dynamics described by Equation (2) can be formulated as follows:

$$\begin{cases} \dot{x}_1 = bi_q + x_2, \\ \dot{x}_2 = \dot{d}_\omega, \end{cases} \quad (8)$$

where $b = \frac{3n_p\Phi}{2J_m}$.

For the above mentioned equation, the ESO can be formulated to estimate the state variables and the lumped disturbance. Its design is presented as follows:

$$\begin{cases} \dot{z}_1 = bi_q + z_2 - \beta_1(z_1 - x_1), \\ \dot{z}_2 = -\beta_2(z_1 - x_1), \end{cases} \quad (9)$$

where z_1 and z_2 are the estimations of x_1 and x_2 . According to the bandwidth selection method [43], $\beta_1 = 2\omega_0$ and $\beta_2 = \omega_0^2$, where $\omega_0 > 0$ is the bandwidth of the ESO.

Theorem 2. Considering the dynamic (2) and its corresponding ESO (9), by selecting the parameters $\beta_1 = 2\omega_0$, $\beta_2 = \omega_0^2$ ($\omega_0 > 0$), the estimation error e_ω in any time is bounded. In addition, increasing the bandwidth of the ESO ω_0 can reduce the estimation error and increase the convergence speed.

Proof. From (8) and (9), the dynamic of the observation error related to the ESO can be formulated as

$$\dot{e}_\omega = Ae_\omega + D\dot{d}_\omega, \quad (10)$$

where $e_\omega = x - z$, $A = \begin{bmatrix} -\beta_1 & 1 \\ -\beta_2 & 0 \end{bmatrix}$, $D = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. The characteristic polynomial of A can be presented in the following form:

$$\lambda(s) = \det(sI - A) = s^2 + \beta_1s + \beta_2, \quad (11)$$

where $\beta_1 = 2\omega_0$, $\beta_2 = \omega_0^2$ ($\omega_0 > 0$). So,

$$\lambda(s) = (s + \omega_0)^2. \quad (12)$$

Thus, the matrix A with the eigenvalues $\lambda_i = -\omega_0 < 0$ ($i = 1, 2$) is a Hurwitz matrix, and the observed error dynamic (10) of the ESO is stable. Additionally,

$$\exp\{At\} = \sum_{i=1}^2 W_i \left[\frac{t^{i-1}}{(i-1)!} \right] \exp\{-\omega_0 t\}, \quad (13)$$

where $W_i \in \mathbb{R}^{2 \times 2}$. It can be found that

$$\|\exp\{At\}\| \leq \sum_{i=1}^2 \|W_i\| \left[\frac{t^{i-1}}{(i-1)!} \right] \exp\{-\omega_0 t\}. \quad (14)$$

Subsequently, for each $\delta > 0$, with $\rho = (\omega_0 - \delta) > 0$, a constant P_1 can be found that satisfies

$$\|\exp\{At\}\| \leq \sum_{i=1}^2 \|W_i\| \left[\frac{t^{i-1}}{(i-1)!} \right] \exp\{-\omega_0 t\} \leq P_1 \exp\{-\rho t\}. \quad (15)$$

Because $\zeta_0 = \sup_{t>0} \|\dot{d}_\omega\|$, the observed error dynamic (10) satisfies

$$\|e_\omega(t)\| \leq P_1 \exp\{-\rho(t-t_0)\} \left[\|e_\omega(t_0)\| - \frac{\zeta_0}{\rho} \right] + \frac{P_1 \zeta_0}{\rho}, t \geq t_0. \quad (16)$$

And

$$\lim_{t \rightarrow \infty} \|e_\omega(t)\| = \frac{P_1 \zeta_0}{\rho}. \quad (17)$$

This indicates that the estimated state z converges exponentially to a bounded region centered around the system state x . Additionally, it can be seen that an increase in the ESO bandwidth ω_0 leads to a higher convergence speed and a reduced radius of the region within which the estimation error remains bounded. This completes the proof. $\square$

3.3. Design of Proposed Controller

The speed error is defined as $e = \omega^* - \omega$, where ω^* is the reference speed. Define the PID sliding mode surface as

$$s = \dot{e} + \rho_1 e + \rho_2 \int_0^t e(\tau) d\tau + d_s, \quad (18)$$

where $\rho_1 > 0$, $\rho_2 > 0$ and $d_s = d_\omega - z_2$.

Assumption 2. The residual disturbance and its derivative are bounded, which are defined by $\varepsilon = \sup_{t>0} \|d_s\|$ and $\eta = \sup_{t>0} \|\dot{d}_s\|$.

The devirative of the global PID surface is

$$\begin{aligned} \dot{s} &= \ddot{e} + \rho_1 \dot{e} + \rho_2 e + \dot{d}_s \\ &= (\ddot{\omega}^* - \ddot{\omega}) + \rho_1 \dot{e} + \rho_2 e + \dot{d}_s. \end{aligned} \quad (19)$$

The controller can be obtained with the ITSMRL (5) and the lumped disturbance observed by the ESO (9):

$$i_q^* = \frac{1}{b} \left\{ \int \left[\rho_1 \dot{e} + \rho_2 e + k_1 |x_1|^{1+\beta} |s|^{1-\beta} \text{sign}(s) + k_2 |s|^{1+\beta} \text{sign}(s) \right] - z_2 \right\} \quad (20)$$

Theorem 3. Consider the PMSM system with uncertainties (2). Utilizing the proposed PID sliding mode surface (18), in cases where the control input is configured as (20), the sliding variable s will asymptotically converge to the bounded region $\min(\Theta_1, \Theta_2)$ in a finite time, where

$$\Theta_1 = \left(\frac{\rho_1 \varepsilon + \eta}{k_2} \right)^{\frac{1}{1+\beta}}, \Theta_2 = \left(\frac{\rho_1 \varepsilon + \eta}{k_1} \right)^{\frac{1}{1-\beta}}.$$

Proof. Consider a Lyapunov function $V = \frac{1}{2}s^2$. By differentiating V with respect to time under the control input (20),

$$\begin{aligned} \dot{V} &= s\dot{s} \\ &= s[\rho_1 \dot{d}_s + \dot{d}_s - k_1 |x_1|^{1+\beta} |s|^{1-\beta} \text{sign}(s) - k_2 |s|^{1+\beta} \text{sign}(s)] \\ &= \rho_1 \dot{d}_s s + \dot{d}_s s - k_1 |x_1|^{1+\beta} |s|^{2-\beta} - k_2 |s|^{2+\beta} \\ &\leq \rho_1 \varepsilon s + \eta s - k_1 |x_1|^{1+\beta} |s|^{2-\beta} - k_2 |s|^{2+\beta} \\ &= -(k_2 |s|^\beta - \frac{\rho_1 \varepsilon + \eta}{s}) s^2 - k_1 |x_1|^{1+\beta} |s|^{2-\beta}. \end{aligned} \quad (21)$$

If $k_2 |s|^\beta - \frac{\rho_1 \varepsilon + \eta}{s} > 0$, there exist $\lambda_1 > 0$ and $\lambda_2 > 0$, which satisfy

$$\begin{aligned} \dot{V} &\leq -\lambda_1 s^2 - \lambda_2 |s|^{2-\beta} \\ &= -2\lambda_1 V - 2^{\frac{2-\beta}{2}} \lambda_2 V^{\frac{2-\beta}{2}}. \end{aligned} \quad (22)$$

The region of s needs to meet $k_2 |s|^\beta - \frac{\rho_1 \varepsilon + \eta}{s} > 0$, as

$$|s| > \left(\frac{\rho_1 \varepsilon + \eta}{k_2} \right)^{\frac{1}{1+\beta}} = \Theta_1. \quad (23)$$

Therefore, the system reaches the following region within a finite time

$$|s| \leq \Theta_1. \quad (24)$$

$\dot{V}$ also can be written as

$$\begin{aligned} \dot{V} &= s\dot{s} \\ &\leq \rho_1 \varepsilon s + \eta s - k_1 |x_1|^{1+\beta} |s|^{2-\beta} - k_2 |s|^{2+\beta} \\ &= -k_2 |s|^\beta |s|^2 - (k_1 |x_1|^{1+\beta} - \frac{\rho_1 \varepsilon + \eta s}{|s|^{2-\beta}}) |s|^{2-\beta}. \end{aligned} \quad (25)$$

If $k_1 |x_1|^{1+\beta} - \frac{\rho_1 \varepsilon + \eta s}{|s|^{2-\beta}} > 0$, there exist $\lambda_3 > 0$ and $\lambda_4 > 0$, which satisfy

$$\begin{aligned} \dot{V} &\leq -\lambda_3 s^2 - \lambda_4 |s|^{2-\beta} \\ &= -2\lambda_3 V - 2^{\frac{2-\beta}{2}} \lambda_4 V^{\frac{2-\beta}{2}}. \end{aligned} \quad (26)$$

The region of s needs to meet $k_1 |x_1|^{1+\beta} - \frac{\rho_1 \varepsilon + \eta s}{|s|^{2-\beta}} > 0$, as

$$|s| > \left(\frac{\rho_1 \varepsilon + \eta}{k_1} \right)^{\frac{1}{1-\beta}} = \Theta_2. \quad (27)$$

Therefore, the system approaches the following region in a finite time:

$$|s| \leq \Theta_2. \quad (28)$$

In conclusion, the sliding variable s will asymptotically converge to the bounded region $\min(\Theta_1, \Theta_2)$ in a finite time, where $\Theta_1 = \left(\frac{\rho_1 \varepsilon + \eta}{k_2}\right)^{\frac{1}{1+\beta}}$, $\Theta_2 = \left(\frac{\rho_1 \varepsilon + \eta}{k_1}\right)^{\frac{1}{1-\beta}}$. This completes the proof. $\square$

4. Simulations and Experiments

In this section, simulations and experiments are implemented to evaluate the performance of this method. The parameters of the PMSM are shown in Table 1.

Table 1. The parameters of the PMSM.

Parameters	Label	Value
rated power	P_{rated}	64 W
rated voltage	V_{rated}	24 V
rated current	I_{rated}	4 A
rated speed	ω_{rated}	3000 rpm
rated torque	τ_{rated}	0.2 N · m
pole pairs	n_p	4
stator inductance	L	0.295 mH
stator resistance	R_s	0.51 Ω
inertia constant	J_m	0.000028 Kg · m ²

To validate the effectiveness and robustness of the proposed approach, three other speed controllers were tested. The first one is a composite controller with a PID sliding surface (18), TSMRL (4) and ESO (9) (PIDSMC+ TSMRL), where the PID sliding mode surface and the ESO are consistent with the proposed algorithm; only the RL is different. The PIDSMC+TSMRL can be expressed as

$$i_q^* = \frac{2J_m}{3n_p\Phi} \left\{ \int_0^t [\rho_1 \dot{e} + \rho_2 e + k_1 |s|^{1-\beta} \text{sign}(s) + k_2 s] - z_2 \right\}. \quad (29)$$

The second one is the TSMC described in [36] with an ESO (9), represented as

$$i_q^* = \frac{2J_m}{3n_p\Phi} \left\{ c|e|^\alpha \text{sat}(e) + p \int_0^t \text{sign}[\dot{e} + c|e|^\alpha \text{sat}(e)] - z_2 \right\}, \quad (30)$$

where $c > 0$, $0 < \alpha < 1$, $p > 0$, $\text{sat}(e) = 1$ when $e > \Delta e$, $\text{sat}(e) = e$ when $e \leq \Delta e$ and $\text{sat}(e) = -1$ when $e < -\Delta e$. The third tested controller is a conventional SMC(CSMC) [44], represented as

$$i_q^* = \frac{2J_m}{3n_p\Phi} \left\{ \int_0^t [\lambda(-\dot{\omega}) + \eta \text{sign}[\lambda(\omega^* - \omega) - \dot{\omega}]] \right\}, \quad (31)$$

where $\lambda > 0$ and $\eta > 0$.

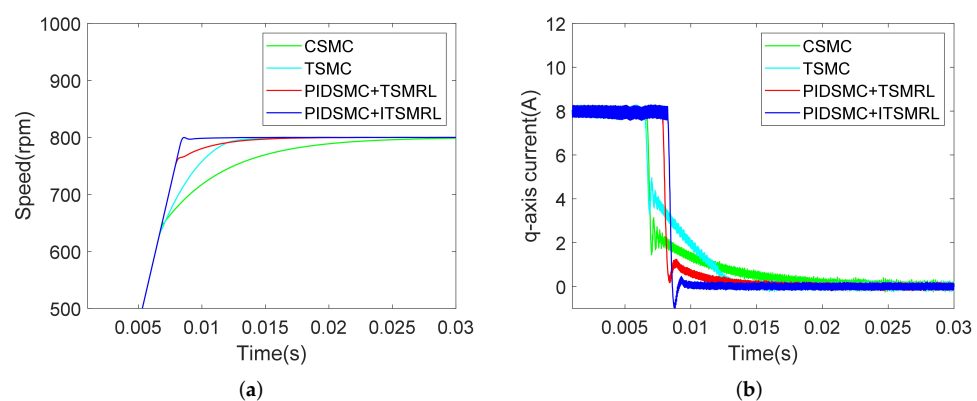
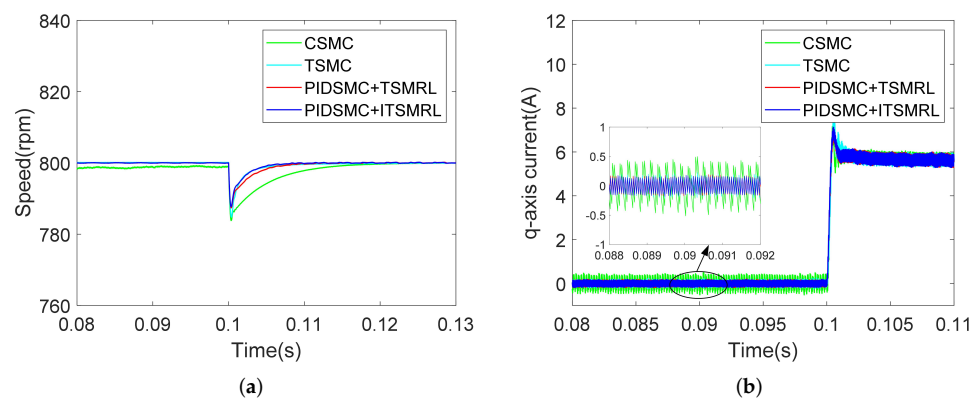
4.1. Simulations

The relevant control parameters used in the experiments are presented in Table 2. To ensure fairness, the controller parameters were selected as the relative optimum after numerous repeated tests, incorporating both dynamic and static performance characteristics. For the surface-mounted PMSM, the parameters of the PI controller in the current loop are designed as $k_{cp} = \alpha L$ and $k_{ci} = \alpha R_s$; here, the bandwidth of the current loop is $\alpha = 2\pi/\tau$. $\tau = L/R_s$ represents the time constant of the PMSM [44].

Table 2. The parameters of controllers.

Algorithms	Parameters
The proposed method	$k_1 = 3.5, k_2 = 160, \rho_1 = 6000, \rho_2 = 0.01, \beta = 0.08, \omega_0 = 10$
PIDSMC+TSMRL	$k_1 = 3.5, k_2 = 380, \rho_1 = 6000, \rho_2 = 0.01, \beta = 0.08, \omega_0 = 10$
TSMC	$c = 1020, p = 25,000,000, \alpha = 0.6, \Delta e = 1, \omega_0 = 10$
CSMC	$\lambda = 200, \eta = 35,000,000$

During the simulation process, the reference speed of the PMSM was 800 rpm when powered on, and a load torque disturbance of $0.2 \text{ N} \cdot \text{m}$ was added at 0.1 s. The speed curve and q -axis current curve of the PMSM during acceleration and loading are shown in Figures 2 and 3, respectively. It can be seen that because of the ITSMRL, compared to the other two controllers, the proposed method has the fastest acceleration, the fastest recovery and the smallest speed fluctuation.

**Figure 2.** Comparison curves at the powering-on phase: (a) speed; (b) q -axis current.**Figure 3.** Comparison curves at the loading phase: (a) speed; (b) q -axis current.

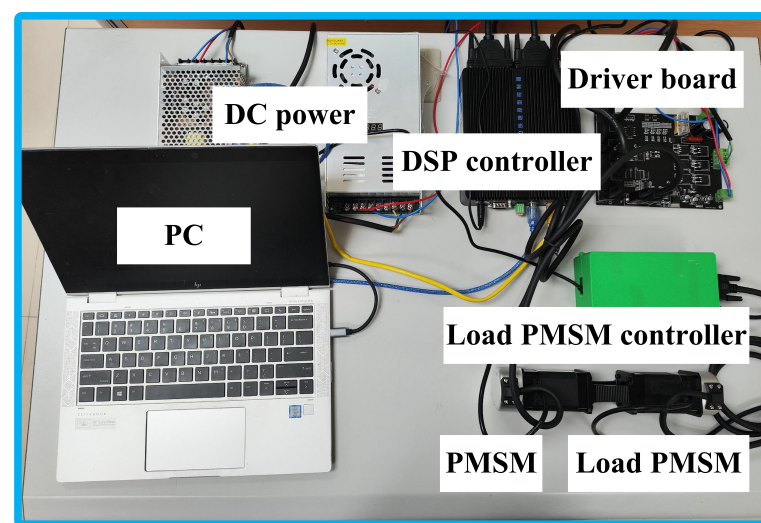
For quantitative comparison, the settling time (ST) and q -axis current root mean square error (RMSEA) were calculated in acceleration simulation, and the recovery time, speed fluctuation and q -axis current root mean square error (RMSEL) were calculated in the loading simulation. The quantitative comparison values are shown in Table 3. It can be seen that the proposed method has the smallest settling time, recovery time, speed fluctuation and q -axis current root mean square error compared to PIDSMC+TSMRL, TSMC and CSMC.

Table 3. Quantitative comparisons of simulation.

Algorithms	ST (s)	RMSEA (A)	RT (s)	SF (rpm)	RMSEL (A)
The proposed method	0.009	0.0853	0.007	12.27	0.7039
PIDSMC+TSMRL	0.014	0.0879	0.011	12.61	0.7142
TSMC	0.013	0.0935	0.011	15.21	0.7326
CSMC	0.025	0.1574	0.015	16.17	0.8057

4.2. Experiments

The acceleration experiments and loading experiments were carried out on the platform shown below to verify the effectiveness and robustness of the proposed method. The hardware platform of the PMSM experiments is shown in Figure 4. The main controller is a Digital Signal Processor of the C2000 series which is manufactured by Texas Instruments (TI) in Dallas, Texas, USA. The load is added by another PMSM.

**Figure 4.** The hardware platform of the PMSM.

To ensure fairness, the controller parameters were optimized through numerous repeated tests, considering both dynamic and static performance. The parameters of the controllers used in the experiments are shown in Table 4. For the surface-mounted PMSM, the PI parameters within the current loop were configured as $k_{cp} = 0.1$ and $k_{ci} = 0.001$.

Table 4. The parameters of the controllers.

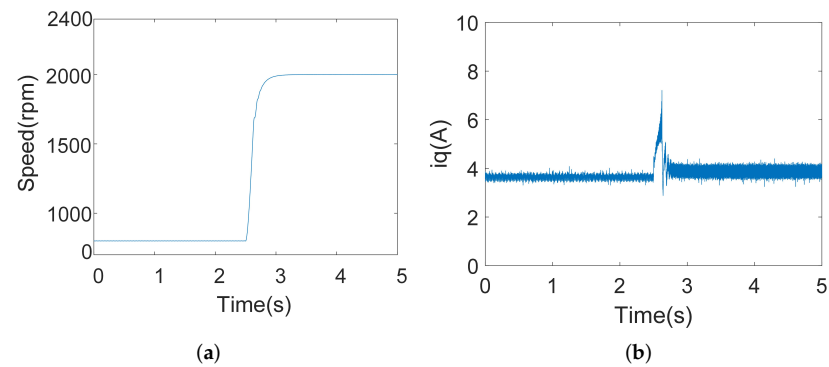
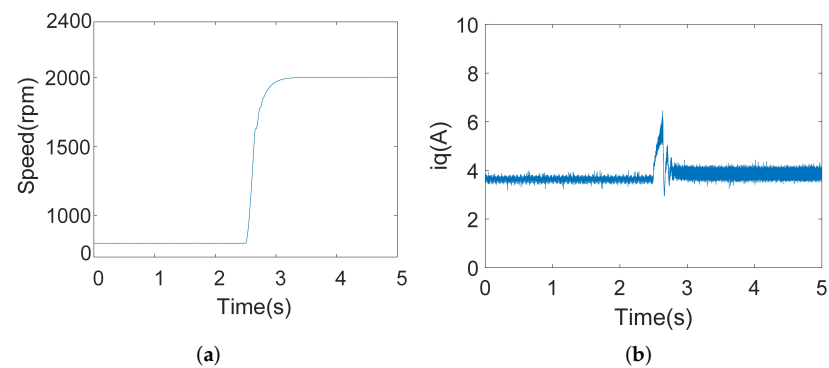
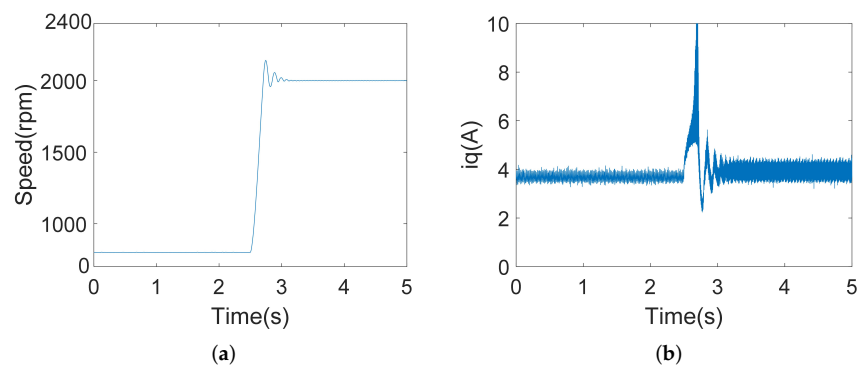
Algorithms	Parameters
The proposed method	$k_1 = 0.001, k_2 = 2.5, \rho_1 = 6000, \rho_2 = 0.01, \beta = 0.08, \omega_0 = 50$
PIDSMC+TSMRL	$k_1 = 0.001, k_2 = 5.5, \rho_1 = 6000, \rho_2 = 0.01, \beta = 0.08, \omega_0 = 50$
TSMC	$c = 2500, p = 8000, \alpha = 0.6, \triangle e = 1, \omega_0 = 50$
CSMC	$\lambda = 500, \eta = 5000$

4.2.1. Acceleration Experiments

In the acceleration experiments, the reference speed of the PMSM was 2000rpm when the PMSM ran at 800 rpm. Figures 5a,b, 6a,b, 7a,b and 8a,b show the speed and i_q response curves under different controllers. The quantitative comparison values are shown in Table 5. It can be seen that the proposed method has the fastest tracking performance compared to PIDSMC+TSMRL, TSMC and CSMC, in addition to reduced chattering, due to the use of the ITSMRL.

Table 5. Quantitative comparison of acceleration experiment.

Algorithms	ST (s)	RMSEA (A)
The proposed method	0.42	0.06
PIDSMC+TSMRL	0.57	0.07
TSMC	0.65	0.09
CSMC	0.73	0.09

**Figure 5.** Acceleration experiment curves under the proposed method: (a) speed; (b) q -axis current.**Figure 6.** Acceleration experiment curves under PIDSMC+TSMRL: (a) speed; (b) q -axis current.**Figure 7.** Acceleration experiment curves under TSMC: (a) speed; (b) q -axis current.

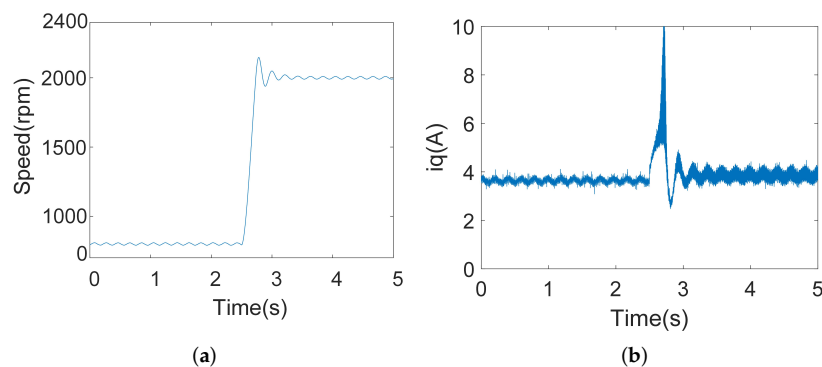


Figure 8. Acceleration experiment curves under CSMC: (a) speed; (b) q -axis current.

4.2.2. Loading Experiments

In the loading experiments, a sudden load torque disturbance was added when the PMSM was running at the speed of 2000 rpm. The response curves for speed and i_q under different controllers are shown in Figures 9a,b, 10a,b, 11a,b and 12a,b. The quantitative comparison values are shown in Table 6. It can be seen that the smallest speed fluctuation and smallest chattering occurred under the proposed method compared to PIDSMC+TSMRL, TSMC and CSMC when loading, as shown in Table 6. Due to the use of the ITSMRL, the proposed method showed reduced chattering and strong robustness.

Table 6. Quantitative comparison of loading experiment.

Algorithms	SF (rpm)	RMSEL (A)
The proposed method	4.5	0.15
PIDSMC+TSMRL	5.9	0.17
TSMC	12.1	0.25
CSMC	69.8	0.23

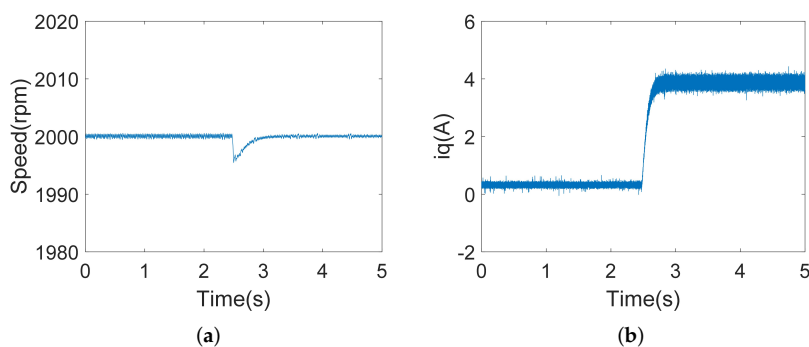


Figure 9. Loading experiment curves under the proposed method: (a) speed; (b) q -axis current.

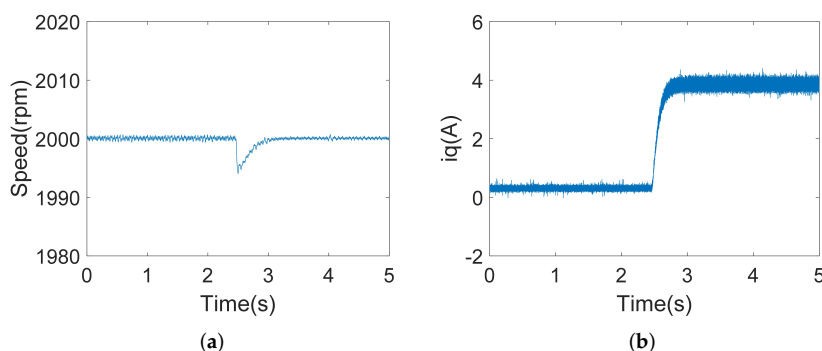


Figure 10. Loading experiment curves under PIDSMC+TSMRL: (a) speed; (b) q -axis current.

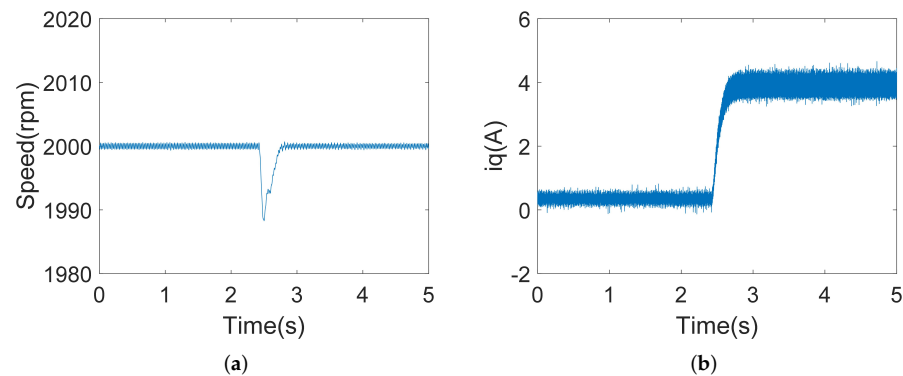


Figure 11. Loading experiment curves under TSMC: (a) speed; (b) q -axis current.

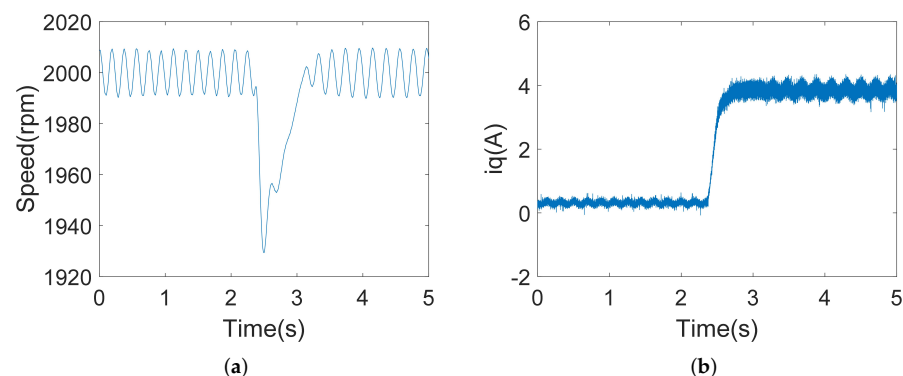


Figure 12. Loading experiment curves under CSMC: (a) speed; (b) q -axis current.

5. Conclusions

A composite controller based on a PID sliding mode surface and the ITSMRL has been proposed in this paper. The ITSMRL has the advantages of fast reaching speed and reduced chattering. Moreover, it is verified that the RL can drive the system to reach the sliding mode surface in a finite time. On the basis of the observed disturbance of the ESO feedforward, a composite controller based on the PID sliding mode surface and ITSMRL is designed to enhance the disturbance rejection ability of the PMSM system. The composite method has the characteristics of fast reaching speed, reduced chattering and strong disturbance rejection ability, and is verified by simulation and experiments.

Author Contributions: G.Q. built the hardware platform of the PMSM and designed the experiment; M.W. wrote the original draft of the manuscript; M.W. and G.C. derived the algorithm; and Q.W. and Y.L. reviewed and edited the manuscript. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by National Natural Science Foundation of China Grant Number: 52307226, Jiaxing Vocational and Technical College Scientific Research Project Grant Number jzyy202415, Teaching Reform Projects of Higher Vocational Education in Zhejiang Province Grant Number jg20240355 and 2024 Jiaxing City Public Welfare Research Program Grant Number 2024AY10034.

Data Availability Statement: The data presented in this study are available on request from the corresponding author.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Kang, W.; Li, H. Improved Sliding Mode Observer Based Sensorless Control for PMSM. *IEICE Electron. Express* **2017**, *14*, 20170934. <https://doi.org/10.1587/elex.14.20170934>.
2. Zerdali, E.; Rivera, M.; Wheeler, P. A Review on Weighting Factor Design of Finite Control Set Model Predictive Control Strategies for AC Electric Drives. *IEEE Trans. Power Electron.* **2024**, *39*, 9967–9981. <https://doi.org/10.1109/TPEL.2024.3370550>.
3. Wang, M.; Liu, Y.; Wang, Q.; Wheeler, P. Current-constraint Speed Regulation for PMSM Based on Port-controlled Hamiltonian Realization and Deep Deterministic Policy Gradient. *IEICE Electron. Express* **2024**, *21*, 20230516.
4. Guo, T.; Sun, Z.; Wang, X.; Li, S.; Zhang, K. A Simple Current-Constrained Controller for Permanent-Magnet Synchronous Motor. *IEEE Trans. Ind. Inform.* **2019**, *15*, 1486–1495. <https://doi.org/10.1109/TII.2018.2860968>.
5. Zhang, N.; Guo, K.; Guo, Y. Research on Low Speed Operation of PMSM Based on Improved Non-Singular Fast Terminal SMC. *IEEE Trans. Appl. Supercond.* **2024**, *34*, 5209705. <https://doi.org/10.1109/TASC.2024.3468070>.
6. Rubio, F.; Pereda, J.; Mora, A.; Rojas, F. Optimal Switching Sequence MPC of a Three-Port-Converter for Variable-Speed PMSM with Hybrid Energy Storage. *IEEE Trans. Power Electron.* **2025**, *40*, 5069–5079. <https://doi.org/10.1109/TPEL.2024.3521504>.
7. Tian, M.; Wang, B.; Yu, Y.; Dong, Q.; Xu, D. Enhanced One Degree-of-Freedom ADRC with Sampled-Data Iterative Learning Controller for PMSM Uncertain Speed Fluctuations Suppression. *IEEE Trans. Transp. Electrification* **2024**, *10*, 8321–8335. <https://doi.org/10.1109/TTE.2024.3351097>.
8. Zou, W.; Shi, T.; Guo, J.; Xiang, Z. A Novel Adaptive Fuzzy Control Scheme for a Class of Nonlinear Planar Systems Under State Constraints. *IEEE Trans. Circuits Syst. II Express Briefs* **2024**, *71*, 827–831. <https://doi.org/10.1109/TCSII.2023.3314751>.
9. Carlet, P.G.; Favato, A.; Bolognani, S.; Dörfler, F. Data-Driven Continuous-Set Predictive Current Control for Synchronous Motor Drives. *IEEE Trans. Power Electron.* **2022**, *37*, 6637–6646. <https://doi.org/10.1109/TPEL.2022.3142244>.
10. Li, S.; Xu, Y.; Zhang, W.; Zou, J. A Novel Two-Phase Mode Switching Control Strategy for PMSM Position Servo Systems with Fast-Response and High-Precision. *IEEE Trans. Power Electron.* **2023**, *38*, 803–815. <https://doi.org/10.1109/TPEL.2022.3200969>.
11. Wang, Y.; Fang, S.; Hu, J.; Huang, D. Multiscenarios Parameter Optimization Method for Active Disturbance Rejection Control of PMSM Based on Deep Reinforcement Learning. *IEEE Trans. Ind. Electron.* **2023**, *70*, 10957–10968. <https://doi.org/10.1109/TIE.2022.3225829>.
12. Liu, X.; Yang, H.; Cai, S.; Lin, H.; Yu, F.; Yang, Y. A Novel Finite-Control-Set Model-Free Predictive Current Control Incorporating Event Triggering Mechanism for PMSM Drives. *IEEE Trans. Ind. Appl.* **2025**, *61*, 322–334. <https://doi.org/10.1109/TIA.2024.3472640>.
13. Zhang, X.; Mao, J.; Zhao, F.; Wang, W.; Jia, R. Speed-current Composite Loop SPMSM Control Based on ADR-SMC. *IEICE Electron. Express* **2024**, *21*, 20240088. <https://doi.org/10.1587/elex.21.20240088>.
14. Xu, B.; Zhang, L.; Ji, W. Improved Non-Singular Fast Terminal Sliding Mode Control with Disturbance Observer for PMSM Drives. *IEEE Trans. Transp. Electrification* **2021**, *7*, 2753–2762. <https://doi.org/10.1109/TTE.2021.3083925>.
15. Wang, S. Nonlinear Uncertainty Estimator-Based Robust Control for PMSM Servo Mechanisms with Prescribed Performance. *IEEE Trans. Transp. Electrification* **2023**, *9*, 2535–2543. <https://doi.org/10.1109/TTE.2022.3212671>.
16. Repecho, V.; Waqar, J.B.; Biel, D.; Dòria-Cerezo, A. Zero Speed Sensorless Scheme for Permanent Magnet Synchronous Machine Under Decoupled Sliding-Mode Control. *IEEE Trans. Ind. Electron.* **2022**, *69*, 1288–1297. <https://doi.org/10.1109/TIE.2021.3062260>.
17. Zhang, L.; Chen, Z.; Yu, X.; Yang, J.; Li, S. Sliding-Mode-Based Robust Output Regulation and Its Application in PMSM Servo Systems. *IEEE Trans. Ind. Electron.* **2023**, *70*, 1852–1860. <https://doi.org/10.1109/TIE.2022.3163536>.
18. Guo, X.; Huang, S.; Lu, K.; Peng, Y.; Wang, H.; Yang, J. A Fast Sliding Mode Speed Controller for PMSM Based on New Compound Reaching Law with Improved Sliding Mode Observer. *IEEE Trans. Transp. Electrification* **2023**, *9*, 2955–2968. <https://doi.org/10.1109/tte.2022.3213562>.
19. Liu, Y.C.; Laghrouche, S.; Depernet, D.; Djerdir, A.; Cirrincione, M. Disturbance-Observer-Based Complementary Sliding-Mode Speed Control for PMSM Drives: A Super-Twisting Sliding-Mode Observer-Based Approach. *IEEE J. Emerg. Sel. Top. Power Electron.* **2021**, *9*, 5416–5428. <https://doi.org/10.1109/jestpe.2020.3032103>.
20. Lu, W.; Zhang, Z.; Wang, D.; Lu, K.; Wu, D.; Ji, K.; Guo, L. A New Load Torque Identification Sliding Mode Observer for Permanent Magnet Synchronous Machine Drive System. *IEEE Trans. Power Electron.* **2020**, *35*, 1156. <https://doi.org/10.1109/tpe.2019.2942788>.
21. Cheng, M.B.; Radisavljevic, V.; Su, W.C. Sliding Mode Boundary Control of a Parabolic PDE System with Parameter Variations and Boundary Uncertainties. *Automatica* **2011**, *47*, 381–387. <https://doi.org/10.1016/j.automatica.2010.10.045>.
22. Zhu, P.; Chen, Y.; Li, M.; Zhang, P.; Wan, Z. Fractional-Order Sliding Mode Position Tracking Control for Servo System with Disturbance. *ISA Trans.* **2020**, *105*, 269–277. <https://doi.org/10.1016/j.isatra.2020.05.032>.
23. Ma, Y.; Li, D.; Li, Y.; Yang, L. A Novel Discrete Compound Integral Terminal Sliding Mode Control with Disturbance Compensation For PMSM Speed System. *IEEE/ASME Trans. Mechatronics* **2022**, *27*, 549–560. <https://doi.org/10.1109/tmech.2021.3068192>.

24. Xu, W.; Junejo, A.; Liu, Y.; Islam, M.R. Improved Continuous Fast Terminal Sliding Mode Control with Extended State Observer for Speed Regulation of PMSM Drive System. *IEEE Trans. Veh. Technol.* **2019**, *68*, 10465–10476. <https://doi.org/10.1109/tvt.2019.2926316>.
25. Qu, L.; Qiao, W.; Qu, L. An Extended-State-Observer-Based Sliding-Mode Speed Control for Permanent-Magnet Synchronous Motors. *IEEE J. Emerg. Sel. Top. Power Electron.* **2021**, *9*, 1605–1613. <https://doi.org/10.1109/jestpe.2020.2990442>.
26. Liu, Y.C.; Laghrouche, S.; Depernet, D.; N'Diaye, A.; Djerdir, A.; Cirrincione, M. Disturbance-observer-based Speed Control for SPMSM Drives Using Modified Super-Twisting Algorithm and Extended State Observer. *Asian J. Control* **2024**, *26*, 1089–1102. <https://doi.org/10.1002/asjc.3270>.
27. Gao, W.; Hung, J. Variable Structure Control of Nonlinear Systems: A New Approach. *IEEE Trans. Ind. Electron.* **1993**, *40*, 45–55. <https://doi.org/10.1109/41.184820>.
28. Ma, H.; Li, Y.; Xiong, Z. Discrete-Time Sliding-Mode Control with Enhanced Power Reaching Law. *IEEE Trans. Ind. Electron.* **2019**, *66*, 4629–4638. <https://doi.org/10.1109/TIE.2018.2864712>.
29. Wang, Y.; Zhu, Y.; Zhang, X.; Tian, B.; Wang, K.; Liang, J. Antidisturbance Sliding Mode-Based Deadbeat Direct Torque Control for PMSM Speed Regulation System. *IEEE Trans. Transp. Electr.* **2021**, *7*, 2705–2714. <https://doi.org/10.1109/TTE.2021.3083074>.
30. Guo, X.G.; Ahn, C.K. Adaptive Fault-Tolerant Pseudo-PID Sliding-Mode Control for High-Speed Train with Integral Quadratic Constraints and Actuator Saturation. *IEEE Trans. Intell. Transp. Syst.* **2021**, *22*, 7421–7431. <https://doi.org/10.1109/TITS.2020.3002550>.
31. Ventura, J.O.; Morales, D.B.; Oliver, J.P.O.; Quesada, E.S.E. Dynamic Sliding Mode Control with PID Surface for Trajectory Tracking of a Multirotor Aircraft. *IEEE Access* **2023**, *11*, 99878–99888. <https://doi.org/10.1109/ACCESS.2023.3314382>.
32. Wang, Q.; Yu, H.; Wang, M.; Qi, X. An Improved Sliding Mode Control Using Disturbance Torque Observer for Permanent Magnet Synchronous Motor. *IEEE Access* **2019**, *7*, 36691–36701. <https://doi.org/10.1109/ACCESS.2019.2903439>.
33. Gao, P.; Zhang, G.; Ouyang, H.; Mei, L. A Sliding Mode Control with Nonlinear Fractional Order PID Sliding Surface for the Speed Operation of Surface-Mounted PMSM Drives Based on an Extended State Observer. *Math. Probl. Eng.* **2019**, *2019*, 7130232. <https://doi.org/10.1155/2019/7130232>.
34. Gao, P.; Zhang, G.; Ouyang, H.; Mei, L. An Adaptive Super Twisting Nonlinear Fractional Order PID Sliding Mode Control of Permanent Magnet Synchronous Motor Speed Regulation System Based on Extended State Observer. *IEEE Access* **2020**, *8*, 53498–53510. <https://doi.org/10.1109/ACCESS.2020.2980390>.
35. Yang, L.; Yang, J. Nonsingular Fast Terminal Sliding-Control for Nonlinear Dynamical Systems. *Int. J. Robust Nonlinear Control* **2011**, *21*, 1865–1879. <https://doi.org/10.1002/rnc.1666>.
36. Tian, M.; Wang, T.; Yu, Y.; Dong, Q.; Wang, B.; Xu, D. Integrated Observer-Based Terminal Sliding-Mode Speed Controller for PMSM Drives Considering Multisource Disturbances. *IEEE Trans. Power Electron.* **2024**, *39*, 7968–7979. <https://doi.org/10.1109/TPEL.2024.3384991>.
37. Dai, C.; Guo, T.; Yang, J.; Li, S. A Disturbance Observer-Based Current-Constrained Controller for Speed Regulation of PMSM Systems Subject to Unmatched Disturbances. *IEEE Trans. Ind. Electron.* **2021**, *68*, 767–775. <https://doi.org/10.1109/TIE.2020.3005074>.
38. Dai, B.; Sun, J.; Yang, J.; Li, S. Dynamic Event-Triggered Disturbance Rejection Control for Speed Regulation of Networked PMSM. *IEEE Trans. Ind. Inform.* **2024**, *20*, 6436–6445. <https://doi.org/10.1109/TII.2023.3342898>.
39. Liu, J.; Yang, J.; Li, S.; Wang, X. Single-Loop Robust Model Predictive Speed Regulation of PMSM Based on Exogenous Signal Preview. *IEEE Trans. Ind. Electron.* **2023**, *70*, 12719–12729. <https://doi.org/10.1109/TIE.2023.3239938>.
40. Wang, Q.; Yu, H.; Li, C.; Lang, X.; Yeoh, S.S.; Yang, T.; Rivera, M.; Bozhko, S.; Wheeler, P. A Low-Complexity Optimal Switching Time-Modulated Model-Predictive Control for PMSM with Three-Level NPC Converter. *IEEE Trans. Transp. Electr.* **2020**, *6*, 1188–1198. <https://doi.org/10.1109/TTE.2020.3012352>.
41. Wang, B.; Tian, M.; Yu, Y.; Dong, Q.; Xu, D. Enhanced ADRC with Quasi-Resonant Control for PMSM Speed Regulation Considering Aperiodic and Periodic Disturbances. *IEEE Trans. Transp. Electr.* **2022**, *8*, 3568–3577. <https://doi.org/10.1109/TTE.2021.3138922>.
42. Zhang, J.; Ren, W.; Sun, X. Extended-State-Observer-Based Nonlinear Control for PMSM Servo Systems with Current Constraints and Voltage Saturations. *IEEE Trans. Transp. Electr.* **2024**, *10*, 2713–2726. <https://doi.org/10.1109/TTE.2023.3304173>.
43. Gao, Z. Scaling and Parameterization Based Controller Tuning. In Proceedings of the Proceedings of the American Control Conference, Denver, CO, USA, 4–6 June 2003.
44. Wang, Y.; Yu, H.; Liu, Y. Speed-Current Single-Loop Control with Overcurrent Protection for PMSM Based on Time-Varying Nonlinear Disturbance Observer. *IEEE Trans. Ind. Electron.* **2022**, *69*, 179–189. <https://doi.org/10.1109/TIE.2021.3051594>.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.